Day 027 | Unit 02 | Teacher plan

Positive Rational Numbers: Fractions and Mixed Numbers

Solve Fraction-Division Stories

Standards: 6.NS.A.1

Learning target: I can write and solve a fraction-division equation for a situation.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Identify the total amount and the size of one group. Use total ÷ group size for the number of groups. A quotient may include a partial group; if the question asks for full servings, use only the whole-number part.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 15

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01 / Learn

Identify the total amount and the size of one group. Use total ÷ group size for the number of groups. A quotient may include a partial group; if the question asks for full servings, use only the whole-number part.

Worked example 1

A ribbon is 1/2 m long. Each piece is 1/4 m. How many piece-lengths are in the ribbon? Include any fractional piece.

1. Identify the total amount and the size of one group.

2. Use total ÷ group size for the number of groups.

3. 1/2 ÷ 1/4 = 1/2 × 4 = 2.

4. State the result in the form requested: 2 piece-lengths.

Answer: 2 piece-lengths

Worked example 2

A jug contains 2/3 L. Each full bottle needs 1/4 L. How many full bottles can be filled, and how much liquid remains?

1. Identify the total amount and the size of one group.

2. 2/3 ÷ 1/4 = 2/3 × 4 = 8/3.

3. Only whole bottles count

4. subtract the liquid used.

Answer: 2 bottles; 1/6 L left

Day 027 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A ribbon is 1/2 m long. Each piece is 1/2 m. How many piece-lengths are in the ribbon? Include any fractional piece.

02 / Constructed response

A ribbon is 1 m long. Each piece is 1/4 m. How many piece-lengths are in the ribbon? Include any fractional piece.

03 / Constructed response

A ribbon is 1/4 m long. Each piece is 1/2 m. How many piece-lengths are in the ribbon? Include any fractional piece.

04 / Constructed response

A ribbon is 1 1/3 m long. Each piece is 5/8 m. How many piece-lengths are in the ribbon? Include any fractional piece.

05 / Constructed response

A ribbon is 1 3/4 m long. Each piece is 2/5 m. How many piece-lengths are in the ribbon? Include any fractional piece.

06 / Constructed response

A ribbon is 1/4 m long. Each piece is 1/3 m. How many piece-lengths are in the ribbon? Include any fractional piece.

07 / Constructed response

A jug contains 2 2/3 L. Each full bottle needs 1/5 L. How many full bottles can be filled, and how much liquid remains?

08 / Model and constructed response

A jug contains 1/2 L. Each full bottle needs 4/3 L. How many full bottles can be filled, and how much liquid remains?

09 / Model and constructed response

A jug contains 2 1/2 L. Each full bottle needs 4/5 L. How many full bottles can be filled, and how much liquid remains?

10 / Constructed response

A jug contains 1 L. Each full bottle needs 2/5 L. How many full bottles can be filled, and how much liquid remains?

Day 027 | Unit 02 | Answer key

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1. 1 piece-lengths

1/2 ÷ 1/2 = 1/2 × 2 = 1.

6.NS.A.1 | Constructed response

2. 4 piece-lengths

1 ÷ 1/4 = 1 × 4 = 4.

6.NS.A.1 | Constructed response

3. 1/2 piece-lengths

1/4 ÷ 1/2 = 1/4 × 2 = 1/2.

6.NS.A.1 | Constructed response

4. 32/15 piece-lengths

4/3 ÷ 5/8 = 4/3 × 8/5 = 32/15.

6.NS.A.1 | Constructed response

5. 35/8 piece-lengths

7/4 ÷ 2/5 = 7/4 × 5/2 = 35/8.

6.NS.A.1 | Constructed response

6. 3/4 piece-lengths

1/4 ÷ 1/3 = 1/4 × 3 = 3/4.

6.NS.A.1 | Constructed response

7. 13 bottles; 1/15 L left

8/3 ÷ 1/5 = 8/3 × 5 = 40/3. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Constructed response

8. 0 bottles; 1/2 L left

1/2 ÷ 4/3 = 1/2 × 3/4 = 3/8. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Model and constructed response

9. 3 bottles; 1/10 L left

5/2 ÷ 4/5 = 5/2 × 5/4 = 25/8. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Model and constructed response

10. 2 bottles; 1/5 L left

1 ÷ 2/5 = 1 × 5/2 = 5/2. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Constructed response